Byung-cheol Kang

Genomics · Bacteriophages · Research Pipelines

A bioinformatics researcher and developer connecting biological questions with analysis design and interpretation. My work extends from microbes to humans, integrating diverse analytical approaches and building research pipelines and databases. AI-powered coding accelerates my research and amplifies my ability to automate workflows and develop analytical tools.

CURRENT AND RECENT ROLES

MicrobiotiX · Senior Researcher · Bioinformatics — 2026.07 - PRESENT

- Analyze bacterial and phage genomes for quality, annotation, taxonomy and host-associated features, with traceable reporting.
- Develop analysis libraries and command-line tools linking host-range matrices with sequence and gene data for cocktail selection, cluster comparison and pan-genome analysis.
- Build RBP/depolymerase reference databases with separate K/KL fields, source citations and experimental/inferred evidence labels.
- Evaluate RBP candidates using sequence, domain and protein-structure evidence and generate candidate sequences and design reports.
- Develop research tools for sequence analysis, primers and cloning; design genome-database workflows and interactive prototypes.

Wittgenbio · Computational Biology Research — 2026.07 - 2026.10

- Built a multi-study CRC scRNA-seq atlas and performed cell-type annotation.
- Cross-checked tumor/normal metadata, author annotations and marker evidence, with versioned label and source tracking.
- Annotated externally supplied normal-tissue atlas data and compared normal-tissue, tumor and patient/specimen expression to assess therapeutic-target candidates.

Past platform evidence. The integrated workflow and operational results in this portfolio describe work performed at Sanigen (2020.03 - 2025.11). They are not presented as MicrobiotiX or Wittgenbio assets or current-company outcomes.

PROJECT 01 · PLATFORM

Integrated Microbial Genomics Platform

PROBLEM

Separate assay scripts, environments, quality checks, server paths, and reporting conventions made analysis difficult to reproduce and review consistently.

SYSTEM DESIGN

Entry point	One Snakemake dispatcher selects 16S, ITS, WGS, or Hybrid profiles.
Execution	Server-specific YAML configuration with Conda, Docker, or Apptainer and pinned BioContainers.
Quality	Input validation, raw-read QC, assembly/analysis checks, review gates, and retained logs.
Delivery	Custom Python/R processing, visualizations, HTML reports, and packaged outputs.

ACTIVE TOOL CHAINS

Profile	Core processing	QC / interpretation
16S / ITS	QIIME 2, DADA2, SILVA classifier, MAFFT, FastTree	FastQC, MultiQC, diversity metrics, custom R/Python reports
WGS	Trimmomatic, SPAdes, Bowtie2, samtools, Prokka	QUAST, BUSCO, Kraken2, Abricate/ CARD/VFDB, eggNOG-mapper
Hybrid	NanoFilt, NanoPlot, Flye or Unicycler, BWA, Pilon, minimap2	QUAST, BUSCO, Prokka, CGView, BLASTn, Abricate, eggNOG-mapper

IMPLEMENTATION AND TECHNICAL REVIEW

- Connected shared analysis, container and server settings with QC, logs and HTML report outputs across the four execution profiles.
- Reran WGS/hybrid assemblies under read-selection and trimming conditions, curated contigs and redelivered corrected reports.
- Corrected reference, FASTA-header and chromosome-contig errors; reran assembly/Prokka annotation and checked circular contigs, graphs and BUSCO results.

EVIDENCE BOUNDARY

Only tools connected to active Snakemake includes and rules are listed as active. Configured or documented but inactive components are not represented as production use.

PROJECT 02 · OPERATIONS

Analysis Operations and Quality Assurance

OPERATING MODEL

Question biological scope → **Execution** workflow + QC → **Delivery** interpretation + provenance

RESPONSIBILITIES AT SANIGEN

- Led NGS analysis services and team operations, including analysis design, hands-on work, result review, SOPs, turnaround management and quality control.
- Performed genomics, microbiome and transcriptome analyses; automated workflows and reports and resolved analysis errors.
- Developed comparative-genomics and k-mer marker-discovery methods and contributed to collaborative NGS pathogen-panel research.
- Contributed to the technology-special listing task force, preparing and revising technical-evaluation and preliminary-listing application materials.
- Delivered custom analyses and method evaluations for external collaborators, client technical support and training materials.
- Maintained HPC/Linux analysis environments, NAS storage and data-transfer infrastructure.

SELECTED ANALYSIS AND DELIVERY OUTPUTS

- Reissued ANI/AMR/virulence/MLST figures, panX SVG/Newick trees, and shotgun/LEfSe subgroup comparisons with KEGG/genus statistics and LDA plots.
- Delivered featureCounts counts, alignment summaries, TSS-depth/TEX comparisons, and RNA target/non-target coverage; compared Nanopore Dorado modes and duplex QC.
- Fixed panel TSV/calculation bugs, tree-JSON/metadata mismatches and missing report nodes; restored eggNOG functional-category, orthology and pathway outputs.
- Resolved NanoPlot dependency/version and missing-visualization errors; corrected QIIME 2 commands and prepared analysis/panX manuals and training.

ADDITIONAL CUSTOM ANALYSES

- Filtered vector/mixed-virus reads, assembled viral reads with metaSPAdes and identified assemblies by BLAST; delivered predicted prophage intervals and mobility evidence.
- Reanalyzed abnormal read-pair/OTU cases and sequence-match tables; provided hybrid-contig VCF/IGV comparisons and pre-NCBI chromosome/contig-submission reviews.
- Verified allergen/accession mappings and custom BLAST target/toxin searches; reconciled NCBI/SILVA/species-label discrepancies with sequence/ANI evidence and documented marker limits.

TRACEABILITY AND INTERPRETATION

Retained communications and analysis outputs support these examples, without a complete workload total. Gene-count delivery is separate from completed differential-expression or enrichment analysis. Platform evaluation and RNA QC do not establish regulatory qualification or therapeutic efficacy.

From Genome Differences to Assay Candidates

COMPARATIVE-GENOMICS MARKER DISCOVERY

Objective. Identify a region that distinguishes a target group while remaining practical for downstream marker or assay development.

Approach. Curate complete genomes, compare whole-genome and pan-genome context, narrow candidate intervals, and review conservation, specificity, and downstream constraints.

Outcome. Prioritized an approximately **17 kb candidate region** for follow-up, with traceable reasoning from genome selection to practical assay constraints.

FOODBORNE-PATHOGEN NGS PANEL

- Developed comparative-genomics and k-mer candidate-marker tools and reviewed panel-analysis code and QA/QC.
- Contributed NGS data analysis and interpretation to collaborative panel development and evaluation.
- Supported interpretation of NGS outputs across food and environmental sample contexts.
- Collaborative outcomes included two peer-reviewed papers in 2023 and one joint patent contribution in 2022.

TECHNICAL-LISTING DOCUMENT CONTRIBUTIONS

- Reviewed diagnostic marker-search/panel functions, panX applicability and algorithm differentiation for technical and patent-review materials.
- Authored, revised and submitted technical-evaluation explanations, figures and tables; wrote the assigned preliminary-listing application section.

BOUNDARIES

> Computational prioritization does not establish diagnostic sensitivity or specificity. Publication and patent records document collaborative outcomes, not sole authorship or universal performance across unevaluated matrices.

SELECTED EVIDENCE

- **Frontiers in Microbiology** 14 (2023): agricultural wastewater pathogen panel.
- **Journal of Microbiology and Biotechnology** 33(1) (2023): fermented-food pathogen panel.
- Joint patent contribution (2022): NGS-based primer set and foodborne-bacteria detection method.

PROJECT 05 · CURRENT ROLE CASE STUDY

Klebsiella Phage Genomics and Tail-RBP Design

GENOME EVIDENCE WORKFLOW

Host + phage genomes
QC + annotation

→

Independent evidence
taxonomy + lifestyle + termini

→

Tail interpretation
KL target + structure

CONTRIBUTION AND EVIDENCE

- Integrated CheckV, Pharokka, Phold, BACPHLIP, PhaTYP, PhaGCN, and PhageTerm outputs while preserving the evidence behind each conclusion.
- Built a tail/RBP and KL-type reference resource with source traceability, confidence labels, and experimentally supported targets explicitly flagged.
- Cross-validated candidate depolymerases with DposFinder, SpikeHunter, DepoScope, sequence search, domain boundaries, and protein-structure evidence.
- Used ProstT5 and Foldseek with experimental PDB references to identify structurally conserved proteins at sequence identities below ordinary search sensitivity.

ANONYMIZED CASE-STUDY RESULTS

- Four genomes converged on lytic **Przondovirus** classifications and short DTR packaging with 180-183 bp terminal repeats.
- Three capsule targets matched experimentally supported references at 95.4-98.8% sequence identity; unsupported cases remained unresolved rather than receiving forced labels.
- One design candidate passed genus, attachment, target-label, experimental-structure, and predicted-structure criteria; its 129-aa anchor alignment spanned 99.2% of the region.

RESEARCH SOFTWARE AND DEVELOPER INTEGRATION

- Separated maximum-coverage/minimum-combination host-range logic; corrected matching, cutoffs, solver time limits and feasible solutions, with independent MILP cross-checks.
- Implemented cluster/defense/pan-genome contracts, JSON adapters and version/error handling; checked source, installed wheels and reproducibility bundles.
- Implemented strand-aware annotation/primer/cloning workflows; repaired input crashes, capsid-capacity reference/margin errors and GenBank/repeat parsing with regression checks.
- Delivered target-specific donor/module ranking and tandem candidates, developer call specifications and reproducible cases, and genome-database QC/review workflow specifications with interactive screen prototypes.

DESIGN OUTPUT AND BOUNDARY

The design bench supports domain swaps, full-module replacements, and tandem additions, with genome FASTA, module multi-FASTA, GenBank, and protein FASTA export. Host K-locus assignments, predicted RBP folding, adsorption, host range, and lytic activity still require bacterial-genome and wet-lab validation.

PROJECT 06 · WITTGENBIO RESEARCH

CRC Atlas and Cell-Type Annotation

CRC SINGLE-CELL ATLAS WORKFLOW

Single-cell datasets
QC + metadata

→

Integrated CRC atlas
harmonization + clustering

→

Cell-type annotation
labels + validation

MULTI-STUDY DATA CURATION AND ANNOTATION

- Standardized public-study metadata, performed cell-level QC, integrated studies, and clustered cells for CRC atlas annotation.
- Cross-checked original author labels and marker evidence against tumor/normal metadata; retained source provenance and version-specific label decisions.
- Produced integrated/annotated data, marker/annotation summaries and version-comparison tables.

TUMOR AND NORMAL-TISSUE EXPRESSION EVIDENCE

CRC atlas
tumor expression

→

Target candidate review
patient / specimen evidence

←

External normal atlas
tissue expression

- Annotated externally supplied normal-tissue atlas data and reviewed tissue-specific expression.
- Combined tissue-specific expression with tumor and patient/specimen evidence to evaluate therapeutic-target candidates.
- Summarized candidate expression, coverage and interpretation limits in comparison tables with follow-up validation items; reviewed raw-count availability and depth differences.

PUBLIC EVIDENCE BOUNDARY

These analyses support RNA-level candidate prioritization. Transcript expression does not establish cell-surface protein abundance, clinical safety, or treatment response. Private datasets and internal identifiers are not disclosed.

PROJECT 07 · INDEPENDENT R&D PROTOTYPE

Phage Lifestyle Evidence Prototype

RESEARCH QUESTION

Can existing tools be combined into an end-to-end system that distinguishes lytic and lysogenic phages while preserving conflicting evidence and uncertainty?

EVIDENCE DESIGN

Quality context	CheckV completeness and contamination context
Annotation	Pharokka and Phold; integrase, repressor, excision, and lysis evidence
Classifiers	BACPHLIP and PhaTYP as independent prediction evidence
Decision layer	Agreement/confidence table with unresolved cases routed to manual review

HOST-RANGE EXTENSION

A related hypothesis-ranking layer considers LPS and outer-membrane gene variation together with phage receptor and annotation evidence. It prioritizes candidates for experiment; it does not prove adsorption, infection, or lytic activity.

PUBLICATION-GRADE NEXT STEPS

- External, leakage-controlled benchmark with known lifestyle labels.
- Comparison against individual tools and transparent simple baselines.
- Sensitivity, specificity, MCC, calibration, and quality-stratified performance.
- Ablation analysis for gene markers, classifiers, and quality context.
- Experimental validation for host-range candidate claims.

CONTACT

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